

D897

Advanced Probability and Statistics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A training firm with 240 employees compares three onboarding formats with 10 new hires in each. The ANOVA table shows $SSB = 60$ and $SSW = 270$. The firm's CEO favors the online format. What is F ?

- A. 3.00
- B. 0.33
- C. 3.33
- D. 0.22

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In a one-way ANOVA, $F = 4$, $MSB = 20$, and the within-groups degrees of freedom are 27. What is SSW ?

- A. 540
- B. 80
- C. 5
- D. 135

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A regional training firm compares four sales-coaching formats. The one-way ANOVA gives $p = 0.01$, and Format C had the highest mean. The trainers ran the study during the busy holiday season. What is the correct conclusion?

- A. Format C is significantly better than each of the other three.
- B. All four format means differ from one another.
- C. The variances of the four formats differ.
- D. At least one format's mean differs; a post hoc test such as Tukey's is needed to say which.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retailer fits $\hat{y} = 12 + 3.5x$, where x is monthly ad spending and $\hat{y}$ is predicted sales, both in thousands of dollars. The correlation is $r = 0.91$, and the data come from its 18 stores. What are predicted sales when ad spending is \$8,000?

- A. \$36,400
- B. \$40,000
- C. \$99,500
- D. \$15,500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A hospital analyst with a decade of experience finds $r = -0.8$ between nurse overtime hours and patient satisfaction scores across 25 units. What share of the variation in satisfaction is explained by the linear relationship with overtime?

- A. 80%
- B. None, because a negative r means the relationship is weak.
- C. 64%
- D. 36%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A regression line has a negative slope and $r^2 = 0.49$. What is r ?

- A. 0.24
- B. -0.49
- C. 0.7
- D. -0.7

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A plant engineer plots machine temperature against defect rate for 150 production runs on the night shift. The scatterplot shows a clear U-shape: defects are high at both low and high temperatures. The correlation is $r = 0.02$. What is the best conclusion?

- A. There is a weak negative relationship.
- B. Temperature and defect rate are unrelated.
- C. There is almost no linear association, but a strong curved relationship exists.
- D. The sample is too small to show any relationship.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An actuary working for a mid-size insurer finds that the number of approved claims in a batch has moment generating function $M(t) = (0.3 + 0.7e^t)^{10}$. The batch was processed on a Monday. What is $E[X]$?

- A. 0.7
- B. 3
- C. 7
- D. 2.1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A random variable has $M'(0) = 4$ and $M''(0) = 20$. What is $\text{Var}(X)$?

- A. 4
- B. 16
- C. 36
- D. 20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A quantitative analyst at a trading firm with offices in two cities proves that a sequence of estimators X_n converges to X in probability. Her proof took three weeks. Which conclusion follows with no further assumptions?

- A. All modes of convergence hold, because they are equivalent.
- B. X_n converges to X in distribution.
- C. X_n converges to X almost surely.
- D. X_n converges to X in mean square.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which classic limit theorem is a statement about convergence in distribution?

- A. The central limit theorem.
- B. None; limit theorems always describe almost-sure convergence.
- C. The strong law of large numbers.
- D. The weak law of large numbers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An analyst using RStudio on a laptop with 16 GB of memory has a data frame `df` with columns `score` (numeric) and `program` (three training programs). She wants to test whether mean score differs by program. Which call fits?

- A. `cor(df$score, df$program)`
- B. `summary(aov(score ~ program, data = df))`
- C. `t.test(score ~ program, data = df)`
- D. `lm(program ~ score, data = df)`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A training firm with 240 employees compares three onboarding formats with 10 new hires in each. The ANOVA table shows $SSB = 60$ and $SSW = 270$. The firm's CEO favors the online format. What is F?

Target word: F statistic **Decides:** $df \text{ between} = 2$, $df \text{ within} = 30 - 3 = 27$.

A	Correct	$MSB = 60/2 = 30$; $MSW = 270/27 = 10$; $F = 30/10 = 3.00$.
B	Inverted operation	MSW/MSB flips the ratio.
C	Wrong stem number	Uses $N = 30$ instead of $N - k = 27$, so $MSW = 9$.
D	Dropped element	$60/270$ divides sums of squares without dividing by degrees of freedom.

2. Answer D Concept 1 • transfer

In a one-way ANOVA, $F = 4$, $MSB = 20$, and the within-groups degrees of freedom are 27. What is SSW ?

Target word: solve for SSW **Decides:** $MSW = MSB / F = 5$.

A	Confusion set	20×27 uses the between-groups mean square.
B	Inverted operation	Multiplies MSB by F instead of dividing.
C	Dropped element	That is MSW ; it still needs to be multiplied by its degrees of freedom.
D	Correct	$MSW = 20/4 = 5$; $SSW = 5 \times 27 = 135$.

3. Answer D Concept 2

A regional training firm compares four sales-coaching formats. The one-way ANOVA gives $p = 0.01$, and Format C had the highest mean. The trainers ran the study during the busy holiday season. What is the correct conclusion?

Target word: what a significant F supports **Decides:** F tests whether any mean differs, not which.

A	Dropped element	Skips the post hoc comparisons that would test each pair.
B	Absolute language	Rejecting "all equal" does not mean every pair differs.
C	Confusion set	Despite its name, ANOVA tests means; it uses variances as its tool.
D	Correct	The omnibus F rejects "all equal" without locating the difference.

4. Answer B Concept 3

A retailer fits $\hat{y} = 12 + 3.5x$, where x is monthly ad spending and $\hat{y}$ is predicted sales, both in thousands of dollars. The correlation is $r = 0.91$, and the data come from its 18 stores. What are predicted sales when ad spending is \$8,000?

Target word: prediction from the line **Decides:** Substitute $x = 8$.

A	Camouflage	Multiplies the prediction by r ; the line already gives the prediction.
B	Correct	$12 + 3.5 \times 8 = 40$ thousand dollars.
C	Inverted operation	$3.5 + 12 \times 8$ swaps the slope and intercept.
D	Dropped element	$12 + 3.5$ leaves out the x value of 8.

5. Answer C Concept 4

A hospital analyst ~~with a decade of experience~~ finds $r = -0.8$ between nurse overtime hours and patient satisfaction scores ~~across 25 units~~. What share of the variation in satisfaction is explained by the linear relationship with overtime?

Target word: r^2 **Decides:** $(-0.8)^2 = 0.64$.

A	Dropped element	Uses $ r $ without squaring it.
B	Confusion set	Sign gives direction; strength comes from the size of r .
C	Correct	Squaring removes the sign: 0.64, or 64%.
D	Inverted operation	$1 - r^2$ is the share left unexplained.

6. Answer D Concept 4 · transfer

A regression line has a negative slope and $r^2 = 0.49$. What is r ?

Target word: solve for r from r^2 **Decides:** $r = -\sqrt{0.49}$ because the slope is negative.

A	Inverted operation	Squares 0.49 again instead of taking its square root.
B	Wrong stem number	Uses r^2 as r .
C	Dropped element	Drops the sign given by the negative slope.
D	Correct	$\sqrt{0.49} = 0.7$, and r takes the slope's negative sign.

7. Answer C Concept 5

A plant engineer plots machine temperature against defect rate ~~for 150 production runs on the night shift~~. The scatterplot shows a clear U-shape: defects are high at both low and high temperatures. The correlation is $r = 0.02$. What is the best conclusion?

Target word: what $r \approx 0$ means **Decides:** r measures only linear association; the plot shows a strong curve.

A	Confusion set	r is slightly positive, and the pattern is strong, not weak.
B	Absolute language	$r \approx 0$ rules out a linear trend, not every relationship.
C	Correct	r captures straight-line trends only; the U-shape is a real pattern.
D	Camouflage	150 runs is enough; the issue is the shape, not the size.

8. Answer C Concept 6

An actuary ~~working for a mid-size insurer~~ finds that the number of approved claims in a batch has moment generating function $M(t) = (0.3 + 0.7e^t)^{10}$. ~~The batch was processed on a Monday~~. What is $E[X]$?

Target word: mean from an MGF **Decides:** The form is binomial with $n = 10$, $p = 0.7$.

A	Dropped element	Leaves out $n = 10$.
B	Confusion set	Uses 0.3, the failure probability, as p .
C	Correct	Binomial(10, 0.7) has mean $np = 7$ (also $M'(0) = 10 \times 0.7$).
D	Adjacent category	$np(1 - p) = 2.1$ is the variance.

9. Answer A Concept 6 · transfer

A random variable has $M'(0) = 4$ and $M''(0) = 20$. What is $\text{Var}(X)$?

Target word: variance from MGF derivatives **Decides:** $\text{Var} = E[X^2] - (E[X])^2$.

A	Correct	$20 - 4^2 = 4$.
B	Wrong stem number	That is $(E[X])^2$, not the variance.
C	Inverted operation	Adds the squared mean instead of subtracting it.
D	Dropped element	$E[X^2]$ alone; the squared mean still has to be subtracted.

10. Answer B Concept 7

A quantitative analyst ~~at a trading firm with offices in two cities~~ proves that a sequence of estimators X_n converges to X in probability. ~~Her proof took three weeks.~~ Which conclusion follows with no further assumptions?

Target word: what convergence in probability implies **Decides:** In probability $\Rightarrow$ in distribution, but not the reverse or upward.

A	Absolute language	The modes are not equivalent; the implications run one way.
B	Correct	Convergence in probability always implies convergence in distribution.
C	Inverted operation	Reverses the chain; almost sure is stronger than in probability.
D	Adjacent category	Mean-square convergence is stronger and needs extra conditions.

11. Answer A Concept 7 · transfer

Which classic limit theorem is a statement about convergence in distribution?

Target word: match a theorem to its mode **Decides:** The CLT describes the limiting distribution of the standardized mean.

A	Correct	The standardized sample mean converges in distribution to a normal.
B	Absolute language	Different theorems use different modes.
C	Confusion set	The strong law is almost-sure convergence of the sample mean.
D	Adjacent category	The weak law is convergence in probability.

12. Answer B Concept 8

An analyst ~~using RStudio on a laptop with 16 GB of memory~~ has a data frame `df` with columns `score` (numeric) and `program` (three training programs). She wants to test whether mean score differs by program. Which call fits?

Target word: the right R function **Decides:** Three group means of a numeric response $\rightarrow$ one-way ANOVA.

A	Confusion set	Correlation needs two numeric variables, not a grouping factor.
B	Correct	<code>aov()</code> fits a one-way ANOVA; <code>summary()</code> gives F and p .
C	Adjacent category	<code>t.test</code> compares exactly two groups; R stops with an error for three.
D	Inverted operation	Puts the grouping variable on the response side of the tilde.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The one-way ANOVA F ratio	$F = MSB / MSW$, where $MSB = SSB / (k - 1)$ and $MSW = SSW / (N - k)$.	Dividing sums of squares directly, or using N instead of $N - k$ for the within-groups degrees of freedom.	Several group means with sums of squares or mean squares given.
2	What a significant ANOVA does and does not say	A significant F says at least one group mean differs; a post hoc test (such as Tukey's) is needed to say which.	Concluding that every mean differs, or that the largest mean beats all others.	"The ANOVA was significant; what can you conclude?"
3	Using and interpreting a regression line	$\hat{y} = b_0 + b_1x$; plug x into the equation to predict, and read b_1 as the predicted change in y for a one-unit increase in x .	Swapping slope and intercept, or adjusting the prediction by r .	A fitted equation and a value of x .
4	r and r^2	r^2 is the share of variation in y explained by the linear model; $r = \pm\sqrt{r^2}$, with the sign of the slope.	Forgetting to square r , or treating a negative r as a weak one.	"What percent of the variation...?"
5	Correlation measures only linear association	r near 0 means no linear association; a strong curved relationship can still exist.	Reading $r \approx 0$ as "no relationship."	A curved or U-shaped scatterplot paired with a small r .
6	Moment generating functions	$M(t) = E[e^{tX}]$; $M'(0) = E[X]$, $M''(0) = E[X^2]$, so $\text{Var}(X) = M''(0) - [M'(0)]^2$; a recognizable form identifies the distribution.	Reporting $E[X^2]$ as the variance, or mixing up p and $1 - p$ in a binomial form.	An $M(t)$ expression or its derivatives at 0.
7	Modes of stochastic convergence	Almost sure $\Rightarrow$ in probability $\Rightarrow$ in distribution; the reverse implications fail in general. The LLN is about probability or almost-sure convergence; the CLT is about distribution.	Reversing the chain, or treating the modes as equivalent.	"Converges in probability / almost surely / in distribution."
8	Matching the R function to the analysis	Compare several group means with <code>aov()</code> (summarized by <code>summary()</code>); model a numeric response with <code>lm(y ~ x)</code> ; two groups with <code>t.test()</code> ; linear association with <code>cor()</code> or <code>cor.test()</code> .	Running a two-group test on three groups, or putting the response on the wrong side of the tilde.	A data frame, a research question, and a list of R calls.

Course scope (public description): Advanced Probability and Statistics provides students with advanced knowledge and skills in probability theory and statistical analysis. Students will learn to conduct an analysis of variance (ANOVA), regression analysis, and correlation analysis. Additionally, they will learn to apply R programming language, generate functions, and use modes of stochastic convergence. The course will focus on applying these concepts to real-world scenarios, decision-making processes, and predictive modeling.